

실습 가이드: OpenFOAM Reacting flow

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Reacting flow

● Governing equations of multiComponentFluid

- A **solver for transient, compressible, turbulent reacting flow** included in the standard OpenFOAM framework.
- Designed to simulate reacting flows with detailed **chemistry, heat release, and turbulence** interactions.
- **Unity Schmidt number (Sc)** and **unity Lewis number (Le)** assumptions are applied for simplification.

$$\text{rhoEqn: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$Sc = \frac{\nu}{D} = \frac{\text{kinematic viscosity}}{\text{mass diffusivity}}$$

$$\text{UEqn: } \frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$Le = \frac{\alpha}{D} = \frac{\text{thermal diffusivity}}{\text{mass diffusivity}}$$

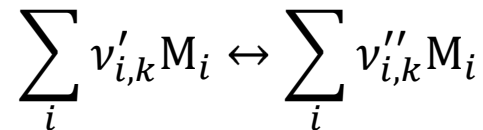
$$\text{YEqn: } \frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\boldsymbol{\mu}_{\text{eff}} \nabla Y_i) + w_i$$

$$\text{EEqn: } \frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) = \frac{\partial p}{\partial t} + \nabla \cdot (\boldsymbol{\alpha}_{\text{eff}} \nabla h) + \sum h_i^0 w_i$$

Reacting flow

● Chemical reaction rates

- Species production rates are calculated from elementary reaction rates.
- Consider elementary reversible reactions



- W_i is molecular weight, $c_j = \rho_j/W_j = Y_j\rho/W_j = X_j p/R^0T$

$$\dot{w}_i = W_i \hat{\omega}_i = W_i \sum_k (\nu''_{i,k} - \nu'_{i,k}) \omega_k$$

$$\omega_k = k_{k,f} \prod_j c_j^{\nu'_{j,k}} - k_{k,b} \prod_j c_j^{\nu''_{j,k}}$$

- The reaction rate constant k is determined by the modified Arrhenius law

$$k = AT^\beta \exp(-E_a/RT)$$

Reacting flow

● Kinetic mechanism

- Chemkin is a software for solving chemical kinetic problems
- Its format is widely used in many simulations of reacting flows
- 1) Gas-phase chemistry file (chem.inp)
 - Elements, Gas phase species, Gas-phase reactions
 - Include the values of A , β , E_a for modified Arrhenius rate coefficient
- 2) Thermodynamic data file (therm.dat)
 - Coefficients of polynomial fits to temperature for species specific heats, enthalpy, and entropy
- 3) Transport data file (tran.dat)
 - Fundamental molecular parameters
 - Lennard-Jones well depth, Lennard-Jones diameter, dipole moment, polarizability, rotational relaxation number
- Example of kinetic mechanism for natural gas combustion – GRI 3.0
 - <http://combustion.berkeley.edu/gri-mech/version30/text30.html>

Reacting flow

● Chemistry ODE integration

- Chemical time scales can be much shorter than flow time scales, resulting in a stiff chemistry system.
- Operator splitting separates transport and chemical reaction contributions.
- An ODE solver integrates the stiff chemistry system in each computational cell.

Chemical reaction ODE system

$$\frac{dc_i}{dt} = \frac{w_i}{W_i}$$

Transport-reaction decomposition

$$\frac{d\phi}{dt} = \textcolor{red}{T}(\phi, x, t) + \textcolor{green}{R}(\phi, t)$$

YEqn: $\frac{\partial \rho Y_i}{\partial t} = \underbrace{-\nabla \cdot (\rho \mathbf{u} Y_i)}_{\text{Transport terms (T)}} + \underbrace{\nabla \cdot (\mu_{\text{eff}} \nabla Y_i)}_{\text{reaction term (R)}} + w_i$

EEqn: $\frac{\partial (\rho h)}{\partial t} = \underbrace{-\nabla \cdot (\rho \mathbf{u} h) - \frac{\partial (\rho K)}{\partial t} - \nabla (\rho \mathbf{u} K) + \frac{\partial p}{\partial t}}_{\text{Transport terms (T)}} + \underbrace{\nabla \cdot (\alpha_{\text{eff}} \nabla h) + \sum h_i^0 w_i}_{\text{reaction term (R)}}$

Reacting flow

● Combustion model

- Combustion model mathematically represents and simplifies **mixing, chemical reactions and heat transfer** so that they can be simulated in CFD.
- Combustion modeling is classified as **Finite rate chemistry model** and **Fast chemistry model**

◆ Fast chemistry ($Da \gg 1$)

- Eddy Dissipation model
- Equilibrium model
- Flamelet-based models (Steady Laminar Flamelet model, Flamelet Generated Manifold model, etc.)

◆ Finite rate chemistry

- Laminar finite rate model
- Eddy Dissipation model/Laminar finite rate model
- **Eddy Dissipation Concept model (EDC)**
- **Partially Stirred Reactor model (PaSR)**
- Transported PDF model

Reacting flow

● Eddy Dissipation Concept (EDC)

- Eddy Dissipation Concept (**EDC**) model is suggested by Magnussen(1981) [*]
- Based on the assumption that chemical reaction occurs in the regions where the dissipation of turbulence energy takes place.

Fine structure length fraction

$$\gamma_L = C_\gamma \left(\frac{\nu \varepsilon}{k^2} \right)^{1/4} \quad \boxed{C_\gamma = 2.1377}$$

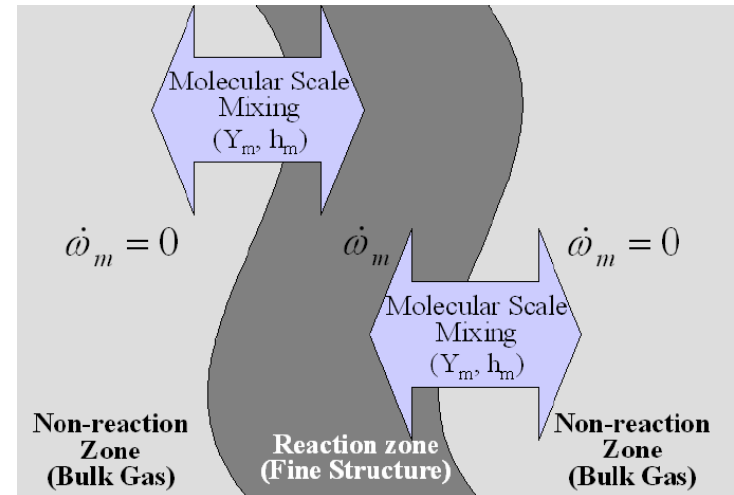
Fine structure residence time

$$\tau^* = C_\tau \left(\frac{\nu}{\varepsilon} \right)^{1/2} \quad \boxed{C_\tau = 0.4083}$$

Mean reaction rate

$$\bar{w}_i = \frac{\gamma_L^{\boxed{2}}}{1 - \gamma_L^{\boxed{2}}} \cdot \frac{\bar{\rho}(\bar{Y}_i - Y_i^*)}{\tau^*}$$

$\boxed{}$: Model coefficients



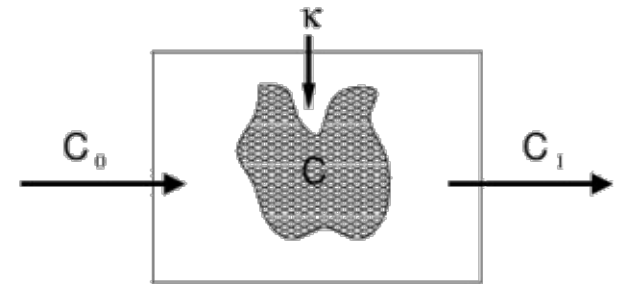
Reacting flow

● Partially-Stirred Reactor (PaSR)

- Partially stirred reactor model is suggested by Chomiak (1990) [*]
- Based on the assumption that reaction zone is perfectly stirred.

Mass fraction of reaction zone

$$\kappa = \frac{\tau_c}{\tau_c + \tau_{mix}}$$



Chemical time scale & turbulent time scale

$$\tau_c = \frac{n_r Y_{tot}^*}{\sum_{n=1}^{n_r} \left(\frac{dY_{n,forward}}{dt} \cdot \nu_{n,sum} \right)} \quad \& \quad \tau_{mix} = \sqrt{\frac{k}{\varepsilon} \left(\frac{\nu}{\varepsilon} \right)^{0.5}} = \boxed{C_{mix}} \sqrt{\frac{\mu_{eff}}{\bar{\rho} \varepsilon}}$$

Mean reaction rate

$$\bar{w}_i = \kappa \cdot \frac{\bar{\rho} (Y_i^* - \tilde{Y}_i)}{\tau_{mix}}$$

$$C_{mix} = \sqrt{\frac{1}{1 + C_\mu Re_t}}$$

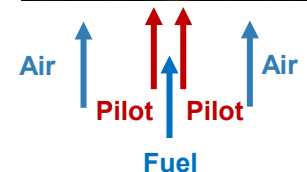
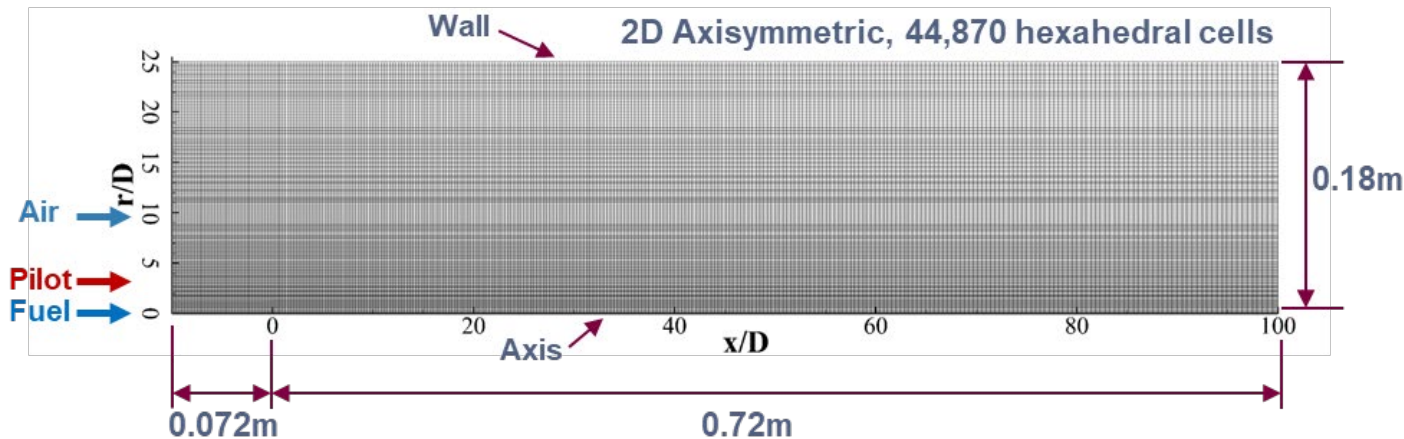
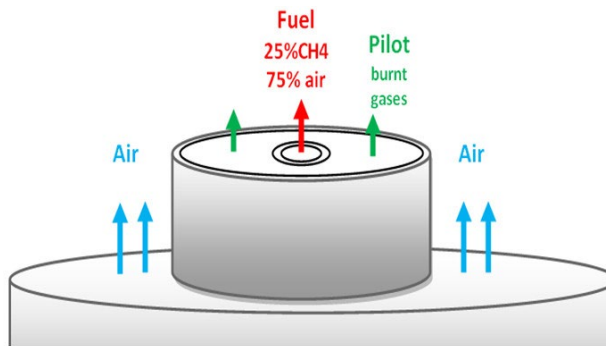
Laminar flow : $C_{mix} = 1$
 Typical turbulent flow : $C_{mix} = 0.001 \sim 0.3$
 Extremely turbulent flow : $C_{mix} = 0$

[*] J. Chomiak, Combustion A Study in theory, fact and application (1990).

Reacting flow

● Tutorial 2: Sandia Flame D [*]

- A piloted partially premixed methane/air diffusion flame
- Widely used for validation of modeling



[*] R.S. Barlow and J. H. Frank, Proc. Combust. Inst. 27:1087-1095(1998)

Reacting flow

● Tutorial 2: Sandia Flame D

- Fuel : CH₄ (25% by vol.) + Air (75% by vol.)
- Pilot : Lean mixture ($\phi = 0.77$)
- Coflow : Dry air

	Fuel	Pilot	Coflow air
U [m/s]	49.6	11.4	0.9
T [K]	294	1880	291
Y_{CH_4}	0.1561	0	0
Y_{CO}	0	4.070E-03	0
Y_{CO_2}	0	1.098E-01	0
Y_{H}	0	2.480E-05	0
Y_{H_2}	0	1.290E-04	0
$Y_{\text{H}_2\text{O}}$	0	9.420E-02	0
Y_{N_2}	0.6473	7.342E-01	0.77
Y_{O}	0	7.470E-04	0
Y_{O_2}	0.1966	5.400E-02	0.23
Y_{OH}	0	2.800E-03	0
Diameter [mm]	7.2	18.2	-

Reacting flow

- Preprocessing: Initial & boundary conditions

Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed
```

```
cd 0
```

```
ll
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed$ cd 0
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed/0$ ll
total 144
drwxr-xr-x  2 ghkrk ghkrk  4096 Oct  5 16:31 ./
drwxr-xr-x 10 ghkrk ghkrk  4096 Oct  6 10:31 ../
-rw-r--r--  1 ghkrk ghkrk 12331 Oct  6 10:30 CH4
-rw-r--r--  1 ghkrk ghkrk  1391 Oct  5 16:31 CO
-rw-r--r--  1 ghkrk ghkrk  1391 Oct  5 16:31 CO2
-rw-r--r--  1 ghkrk ghkrk  1047 Oct  5 16:31 G
-rw-r--r--  1 ghkrk ghkrk  1390 Oct  5 16:31 H
-rw-r--r--  1 ghkrk ghkrk  1391 Oct  5 16:31 H2
-rw-r--r--  1 ghkrk ghkrk  1391 Oct  5 16:31 H2O
-rw-r--r--  1 ghkrk ghkrk 27554 Oct  6 10:30 N2
-rw-r--r--  1 ghkrk ghkrk  1390 Oct  5 16:31 O
-rw-r--r--  1 ghkrk ghkrk 27553 Oct  6 10:30 O2
-rw-r--r--  1 ghkrk ghkrk  1390 Oct  5 16:31 OH
-rw-r--r--  1 ghkrk ghkrk  1485 Oct  6 10:30 T
-rw-r--r--  1 ghkrk ghkrk  1442 Oct  5 16:31 U
-rw-r--r--  1 ghkrk ghkrk  1391 Oct  5 16:31 Ydefault
-rw-r--r--  1 ghkrk ghkrk  1598 Oct  5 16:31 alphas
-rw-r--r--  1 ghkrk ghkrk  1985 Oct  5 16:31 epsilon
-rw-r--r--  1 ghkrk ghkrk  1651 Oct  5 16:31 k
-rw-r--r--  1 ghkrk ghkrk  1680 Oct  5 16:31 nut
-rw-r--r--  1 ghkrk ghkrk  1313 Oct  5 16:31 p
```

Additionally required for reacting flow
(initial conditions of species)

Reacting flow

● Preprocessing: Initial & boundary conditions

Command Line

```
vi N2
```

- Species compositions in OpenFOAM are specified using mass fractions, Y_i
- Repeat this step for every other species with a non-zero mass fraction in the initial internal field or at any inlet boundary.

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       volScalarField;
    location    "0";
    object      N2;
}
// *****

dimensions      [];
internalField    uniform 0.77;

boundaryField
{
    inletCH4
    {
        type      fixedValue;
        value      uniform 0.6473;
    }
    wallOutside
    {
        type      zeroGradient;
    }
    wallTube
    {
        type      zeroGradient;
    }
    inletPilot
    {
        type      fixedValue;
        value      uniform 0.7342;
    }
    inletAir
    {
        type      fixedValue;
        value      uniform 0.77;
    }
}
```

Reacting flow

● Preprocessing: Initial & boundary conditions

Command Line

```
vi Ydefault
```

- Ydefault consisted of zero in initial internal field and all inlet boundary fields.
- When all species contained in kinetic mechanism are not present in the current directory, it serves as a substitute for the remaining absent species.

```
/*----- C++ -----*/
//
// \ V / F i e l d
//  \ / O p e r a t i o n
//   V A n d
//   M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website: https://openfoam.org
// Version: 12
//
/*-----*/

FoamFile
{
    format      ascii;
    class       volScalarField;
    location    "0";
    object      Ydefault;
}
// ***** //

dimensions      [];

internalField    uniform 0;

boundaryField
{
    wallTube
    {
        type      zeroGradient;
    }

    outlet
    {
        type      inletOutlet;
        inletValue $internalField;
    }

    inletPilot
    {
        type      fixedValue;
        value      uniform 0;
    }

    inletAir
    {
        type      fixedValue;
        value      uniform 0;
    }

    wallOutside
    {
        type      zeroGradient;
    }
}
```

Reacting flow

- Preprocessing: Kinetic mechanism

- Chemkin-format Gas-phase chemistry file

Command Line

```
cd ../chemkin
```

```
vi chem.inp
```

Elements

Species

Reactions

```
! GRI-Mech Version 3.0 7/30/99 CHEMKIN-II format
! See README30 file at anonymous FTP site unix.sri.com, directory gri;
! WorldWideWeb home page http://www.me.berkeley.edu/gri_mech/ or
! through http://www.gri.org , under 'Basic Research',
! for additional information, contacts, and disclaimer

ELEMENTS
O H C N AR
END

SPECIES
H2      H      O      O2      OH      H2O      HO2      H2O2
C       CH     CH2     CH2(S) CH3     CH4      CO       CO2
HCO     CH2O    CH2OH   CH3O   CH3OH   C2H      C2H2     C2H3
C2H4    C2H5     C2H6    HCCO   CH2CO   HCCOH    N2       AR
C3H7    C3H8     CH2CHO  CH3CHO
END

REACTIONS
2O+M<=>O2+M 1.200E+17 -1.000 .00
H2/ 2.40/ H2O/15.40/ CH4/ 2.00/ CO/ 1.75/ CO2/ 3.60/ C2H6/ 3.00/ AR/ .83/
O+H+M<=>OH+M 5.000E+17 -1.000 .00
H2/2.00/ H2O/6.00/ CH4/2.00/ CO/1.50/ CO2/2.00/ C2H6/3.00/ AR/ .70/
O+H2<=>H+OH 3.870E+04 2.700 6260.00
O+HO2<=>OH+O2 2.000E+13 .000 .00
O+H2O2<=>OH+HO2 9.630E+06 2.000 4000.00
O+CH<=>H+CO 5.700E+13 .000 .00
O+CH2<=>H+HCO 8.000E+13 .000 .00
O+CH2(S)<=>H2+CO 1.500E+13 .000 .00
O+CH2(S)<=>H+HCO 1.500E+13 .000 .00
O+CH3<=>H+CH2O 5.060E+13 .000 .00
O+CH4<=>OH+CH3 1.020E+09 1.500 8600.00
O+CO(+M)<=>CO2(+M) 1.800E+10 .000 2385.00
```

Reacting flow

- **Preprocessing: Kinetic mechanism**

- Chemkin-format Gas-phase thermodynamic data file

Command Line

```
vi therm.dat
```

```
THERMO ALL
  250.000  1000.000  5000.000
! GRI-Mech Version 3.0 Thermodynamics released 7/30/99
! NASA Polynomial format for CHEMKIN-II
! see README file for disclaimer
0          L 1/900  1          G  200.000  3500.000  1000.000  1
  2.56942078E+00-8.59741137E-05 4.19484589E-08-1.00177799E-11 1.22833691E-15 2
  2.92175791E+04 4.78433864E+00 3.16826710E+00-3.27931884E-03 6.64306396E-06 3
-6.12806624E-09 2.11265971E-12 2.91222592E+04 2.05193346E+00 4
```

Polynomial Coefficients for NASA7

Reacting flow

● Preprocessing: Kinetic mechanism

- The standard multicomponent solver module uses the Sutherland transport model to calculate viscosity (μ_i) and thermal conductivity (λ_i).

Command Line

```
vi transportSutherland
```

$$\mu_i = \frac{A_s \sqrt{T}}{1 + T_s/T}$$

$$\lambda_i = \mu_i c_v \left(1.32 + 1.77 \frac{R}{c_v} \right)$$

```
/*----- C++ -----*/
//
//      F ield      | OpenFOAM: The Open Source CFD Toolbox
//      O peration   | Website: https://openfoam.org
//      A nd         | Version: 12
//      M anipulation|
//
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "chemkin";
    object      transportSutherland;
}
// ***** //

"."
{
    transport
    {
        As 1.512e-06;
        Ts 120.;
    }
}
```

Reacting flow

● Preprocessing: Kinetic mechanism

- Convert Chemkin-format files into OF-format reaction and thermodynamic file

Command Line

```
cd ..
```

```
chemkinToFoam \
```

```
chemkin/chem.inp \  
chemkin/therm.dat \  
chemkin/transportSutherland \
```

Input

```
constant/reactions \  
constant/thermo.STM
```

Output

```
cd constant
```

```
vi reactions
```

```
reactions  
{  
  un-named-reaction-0  
  {  
    type          reversibleThirdBodyArrhenius;  
    reaction       "2O = O2";  
    A              1.2e+11;  
    beta           -1;  
    Ta             0;  
    coeffs
```

Reaction rate type
Reaction
Arrhenius coefficients

Reacting flow

- **Preprocessing: Kinetic mechanism**

- Convert Chemkin-format files into OF-format reaction and thermodynamic file

Command Line

```
vi thermo.STM
```

Species list

```
species 36 ( H2 H O O2 OH H2O HO2 H2O2 C CH CH2 CH2(S) CH3 CH4 CO CO2 HCO CH2O CH2OH CH3O CH3OH C2H C2H2 C2H3 C2H4 C2H5 C2H6 HCCO CH2CO HCCOH N2 AR C3H7 C3H8 CH2CHO CH3CHO );
OH
{
  specie
  {
    molWeight      17.00737;      Molecular weight
  }
  thermodynamics
  {
    Tlow           200;
    Thigh          3500;
    Tcommon        1000;
    highCpCoeffs   ( 3.09288767 0.000548429716 1.26505228e-07 -8.79461556e-11 1.17412376e-14 3858.657 4.4766961 );
    lowCpCoeffs    ( 3.99201543 -0.00240131752 4.61793841e-06 -3.88113333e-09 1.3641147e-12 3615.08056 -0.103925458 );
  }
  transport
  {
    As             1.512e-06;      Sutherland
    Ts             120;             coefficients
  }
  elements
  {
    O              1;
    H              1;
  }
}
```

Reacting flow

- **Preprocessing: Chemistry model**

Command Line

vi chemistryProperties

```

/*-----*- C++ -*-----*/
    F ield      | OpenFOAM: The Open Source CFD Toolbox
    O peration  | Website:  https://openfoam.org
    A nd        | Version:  12
    M anipulation |
/*-----*- C++ -*-----*/

FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       chemistryProperties;
}
// *****

#includeEtc "caseDicts/solvers/chemistry/TDAC/chemistryPropertiesFlame.cfg"

chemistryType
{
    solver      ode;
}

chemistry      on;

initialChemicalTimeStep 1e-7;

odeCoeffs
{
    solver      seulex;
    absTol      1e-8;
    relTol      1e-1;
}

```


Reacting flow

- Preprocessing: Mesh Generation & setFields

Command Line

```
cd ..
```

```
blockMesh
```

```
setFields
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD$ blockMesh
/*-----*
//
=====
\\      F ield      | OpenFOAM: The Open Source CFD Toolbox
\\      O peration  | Website:  https://openfoam.org
\\      A nd        | Version:  12
\\      M anipulation|
//
/*-----*/
Build   : 12-86e126a7bc4d
Exec    : blockMesh
Date    : Oct 06 2026
Time    : 10:38:40
Host    : "20225263-p01"
PID     : 72615
I/O     : uncollated
Case    : /home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// *****
Create time

Deleting polyMesh directory
"/home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD/constant/polyMesh"
Reading "blockMeshDict"

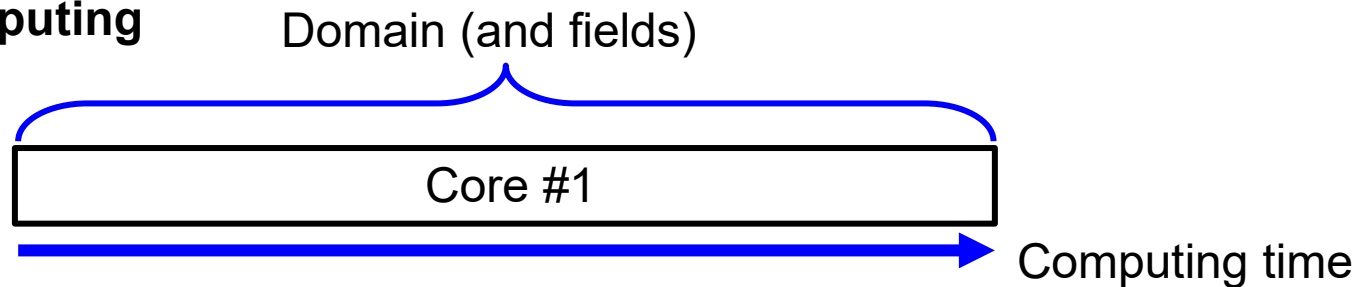
Creating block mesh from
"system/blockMeshDict"
```

Reacting flow

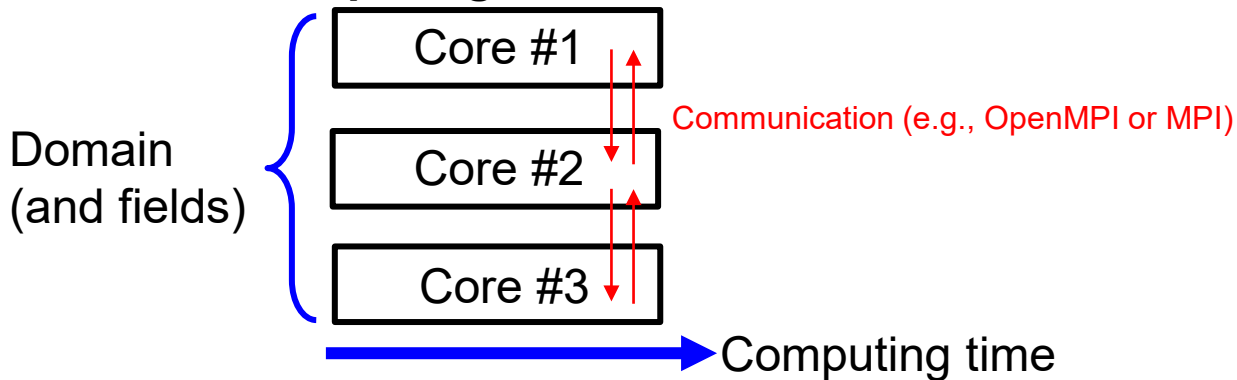
- **Preprocessing: Domain decomposition**

- Parallel computing

1. Serial computing



2. Parallel computing



Reacting flow

- Preprocessing: Domain decomposition
 - Parallel computing

In system/decomposeParDict file

```
/*-----* C++ *-----*/
//
// \ \ \ \ \
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website:  https://openfoam.org
// Version:  12
//
FoamFile
{
    format      ascii;
    class       dictionary;
    note        "mesh decomposition control dictionary";
    object      decomposeParDict;
}
// ***** //

numberOfSubdomains 4;

method            simple;

simpleCoeffs
{
    n              (1 1 4);
    delta          0.001;
}

// ***** //
```

Command Line

decomposePar

Reacting flow

- Preprocessing: Domain decomposition

- Parallel computing

Command Line

11

```
Decomposing volVectorFields
```

```
U
```

```
Decomposing point fields
```

```
(no point fields)
```

```
End
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD$ 11
total 48
drwxr-xr-x 10 ghkrk ghkrk 4096 Oct  6 10:29 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 0/
-rwxr-xr-x  1 ghkrk ghkrk  328 Oct  5 16:31 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  578 Oct  5 16:31 Allrun*
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:29 constant/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor0/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor1/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor2/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  6 10:29 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:29 system/
```

Domain is decomposed

Reacting flow

- Solving: Run the simulation
 - Use multicomponentFluid solver

Command Line

```
mpirun -np 4 foamRun -parallel
```

Number of processors to use
(number of decomposed domain)

Command want to execute

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD$ cd processor0
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD/processor0$ ll
total 52
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 ./
drwxr-xr-x 18 ghkrk ghkrk 4096 Oct  6 10:49 ../
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:37 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:38 1000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:38 2000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:39 3000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:40 4000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:41 5000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:42 6000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:48 7000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8000/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:37 TDAC/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:37 constant/
```

Results is in each processor directory

Reacting flow

● Post-Processing

```
U
Reconstructing surfaceScalarFields

phi
Reconstructing point fields
(no point fields)
Collecting uniform files

End

ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlamed$ ll
total 92
drwxr-xr-x 18 ghkrk ghkrk 4096 Oct  6 10:49 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 1000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 2000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 3000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 4000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 5000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 6000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 7000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8000/
-rwxr-xr-x  1 ghkrk ghkrk  310 Oct  6 10:43 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  552 Oct  6 10:42 Allrun*
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:46 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 constant/
-rw-r--r--  1 ghkrk ghkrk 3342 Oct  6 10:37 log.blockMesh
-rw-r--r--  1 ghkrk ghkrk 1107 Oct  6 10:37 log.chemkinToFoam
-rw-r--r--  1 ghkrk ghkrk 1514 Oct  6 10:37 log.setFields
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor0/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor1/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor2/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:47 system/
```

Command Line

```
reconstructPar
```

```
ll
```

Results of each processor is merged

Reacting flow

● Post-Processing

```
U
Reconstructing surfaceScalarFields

phi
Reconstructing point fields

(no point fields)

Collecting uniform files

End

ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/2.SandiaFlameD$ ll
total 92
drwxr-xr-x 18 ghkrk ghkrk 4096 Oct  6 10:49 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:31 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 1000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 2000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 3000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 4000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 5000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 6000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 7000/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8000/
-rwxr-xr-x  1 ghkrk ghkrk  310 Oct  6 10:43 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  552 Oct  6 10:42 Allrun*
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:46 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:43 constant/
-rw-r--r--  1 ghkrk ghkrk 3342 Oct  6 10:37 log.blockMesh
-rw-r--r--  1 ghkrk ghkrk 1107 Oct  6 10:37 log.chemkinToFoam
-rw-r--r--  1 ghkrk ghkrk 1514 Oct  6 10:37 log.setFields
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor0/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor1/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor2/
drwxr-xr-x 13 ghkrk ghkrk 4096 Oct  6 10:49 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:47 system/
```

Command Line

```
rm -rf processor*
```

Do not need files in processor* after reconstruct

Reacting flow

● Post-Processing

- Do the same process you have done in cavity Tutorial

Command Line

```
touch reacting.foam
```

Copy & paste files
to C: or D: drive

Open foam file
with paraview

